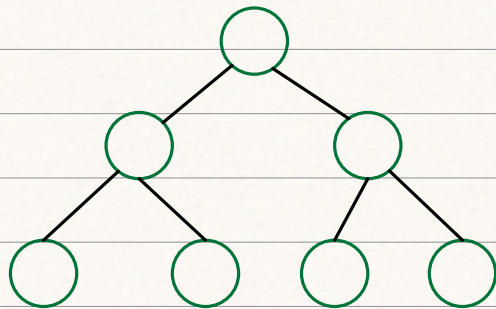


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Best First Search :-

Follow a single path at a time but switch path whenever some competing path looks more promising than the current one does.



Total cost from initial state to goal state $\rightarrow J = g + h \rightarrow$ Additional cost from current node to goal state.
 $\downarrow$
cost from initial state to current node

Best First Search Algo.

- ① Start with OPEN containing only initial
- ② Untill a goal is found or no nodes left in OPEN do:
 - (a) PICK the best node from OPEN
 - (b) Generate its successors
 - (c) For each successor do:
 - (i) if it had not been generated before add it to OPEN and record its parent

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(ii) if it had been generated before change the parent if this new path is more promising.
update cost also.

* Best First Search Algo. is simplest form of A* algo.

⇒ A* algorithm ↪ Hart. et. al (1968-1972)

1. Start with OPEN containing only the initial node

Set node g value to 0.

$$f' = h + 0$$

CLOSED: Empty

2. Until Goal State is found or OPEN list is empty.

- (a) Pick a best node from OPEN with having least value of f'
- (b) if best node is goal state then QUIT
otherwise generate SUCCESSOR of Best node.

(i) Set Successor to point facts to best node

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(ii) Compute $g(\text{successor}) = g(\text{best node})$

+
cost of getting
from best to
successor

(iii) See successor was not on OPEN
but in CLOSED.

Add old to list of best
node successor if new path is
better than old path then
compute g again

(iv) if successor is not in OPEN
& CLOSED then add it to OPEN
, add it to list of best node
successor.